

CHAITANYA N.V.K

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Technical Skills

Languages: Java, Python, Go, JavaScript, TypeScript, C++, SQL, NoSQL
Distributed Systems: Microservices, Kafka, RabbitMQ, REST, gRPC, Event-driven, Multithreading
Cloud & DevOps: AWS (Lambda, EC2, S3, RDS), Kubernetes, Docker, Jenkins, GCP, Azure, CI/CD
Databases: MongoDB, Cassandra, DynamoDB, PostgreSQL, Redis, ElastiCache
Frameworks: Spring Boot, Node.js, Express, React, Angular, TDD
Tools: Git, Linux, Prometheus, Grafana, CloudFormation, JProfiler, JConsole

Experience

Workday

Aug 2025 – Now

Software Development Engineer

Atlanta, GA

- **Zero-Loss DB Migrations Framework:** Designed and implemented a framework enabling developers to perform database migrations and rollbacks with **zero data loss**, increasing system reliability and deployment confidence across services.
- **Automated Rollback Execution:** Developed **automated recovery tooling** that triggers immediate rollback workflows upon failed migrations, protecting data integrity and cutting recovery time from **hours to minutes**.
- **Change Safety Guardrails:** Built verification and compliance checks to ensure schema consistency and backward compatibility during releases, reducing critical production incident risk by **40%**.
- **Developer Enablement:** Standardized migration templates and scripts enabling **100+ developers** to adopt a consistent deployment model across microservices.

Amazon

Sep 2023 – May 2025

Software Development Engineer

Seattle, WA

- **UNFI Enhancement:** Improved UNFI review and deep dive pages using **AWS Lambda**, streamlining vendor item-store updates and reducing decision-making time from **hours to minutes**.
- **WRAP Backfills:** Led backfill initiatives in the UK, JP, and US, enabling online sales for physical stores and **increasing sales by 20%** in the initial months.
- **WRAP Migration:** Optimized migration workflows and **Lambda configurations**, achieving a **30% reduction in migration time**.
- **Guardrails:** Developed workflow rules and validations, resulting in a **60% decrease in user tickets** by automating error handling and data validation.
- **Risks:** Proactively addressed security and software issues, reducing average turnaround time by **two weeks**.
- **Sourcing Grid:** Designed a new **API and user interface** to streamline configuration, simplifying selection from **five attributes to one**.
- **Grocery Listing:** Enhanced the listing process for the supply chain, cutting user story time from **10 minutes to 4 minutes** for over **25 million global users**.

RE IT Services Inc

Feb 2023 – July 2023

Software Development Engineer

Atlanta, GA

- **Data Pipeline:** Implemented auto archival of database records, resulting in a **90% improvement in query performance**.
- **API Scaling:** Enhanced APIs to support increased load and automated tasks within the **CI/CD pipeline** using **Jenkins** and **Docker**, saving **6 hours of manual work per task**.
- **Data Processing:** Developed **event-driven data processing** solutions using **Kafka** to standardize multi-source data across **18 systems**, eliminating data discrepancies and reducing on-call workload by **11 hours per week**.

Amazon

May 2022 – Aug 2022

SDE Intern at AWS

Metro DC, VA

- **AEA Onboarding:** Collaborated to streamline the **AWS Enterprise Access** onboarding process, reducing average onboarding time by **7 hours per instance**.
- **Topos:** Developed a full-stack console using **Python** to address security issues identified by network scans, enhancing security for **9,000 Amazon services**.

Darwinbox (Cloud-based HRMS platform)

Jun 2018 – Aug 2021

Senior Software Engineer

Hyderabad, AP

- **Microservices Migration:** Led the migration to **microservices architecture** using **Node.js** and **Docker**, reducing throughput time from **6 minutes to 1 minute**, and enabling business workflows as a service.

- **Payslip Customization:** Developed a flexible framework for payslip customization, attracting clients with over **1 million employees** to the platform.
- **Dynamic Reporting Framework:** Implemented a PHP-based reporting engine, handling on-demand data requests for **200,000 employees** in a single operation.
- **MongoDB Optimization:** Executed optimization strategies that saved **\$800K in infrastructure costs** by reducing MongoDB usage.
- **Multi-Payroll Integration:** Created **stateless APIs** for payroll systems, serving **1 million users** across multiple systems.
- **Payroll Automation:** Reduced manual processing hours from **40 to 12** by automating payroll workflows using **Redis** and **RabbitMQ**.
- **Bulk Notifications:** Developed bash scripts for bulk email and notification processing, supporting communications for over **200,000 users**.
- **Taxation System:** Built a framework for handling multi-country tax systems, attracting **50 new clients** due to enhanced functionality.
- **Recruitment Automation:** Constructed a recruitment automation pipeline using **event emitters**, reducing hiring time by **60%**.

Darwinbox (Cloud-based HRMS platform)

Dec 2017 – Apr 2018

Intern

Hyderabad, AP

- **Chatbot:** Leveraged **Wit.ai** (Facebook's NLP tool) to automate tasks for over **1 million customers**, streamlining HR workflows using **Go**.
- **OCR Integration:** Integrated **OCR** into the reimbursement module, automating data extraction from bills and reducing processing time by **90%**.

Education

Georgia State University - Full Scholarship along with GTA

Aug. 2021 – Dec. 2022

Masters of Science in Computer Science; GPA:3.96/4.0

Atlanta, Georgia

Vignan University - Merit scholarship

Aug. 2014 – Aug 2018

Bachelor of Technology in Computer Science; GPA: 3.65/4.0

Guntur, AP, India

Personal Projects

Gumo | Node JS, Mongo DB, NPM package

April 2021

- * A web-crawler and scraper to extract data from a family of nested dynamic web pages, with added enhancements to assist knowledge mining applications.

NFL Health & Safety - Helmet segmentation and labeling | Python, Pandas, NumPy

Dec 2021

- * Published the findings as, "Real Time Multi-Object Detection for Helmet Safety". 19 May 2022, Arxiv

Research Publications

- * **Nuthalapati, V.K.C.**, et al., "*Real-Time Multi-Object Detection for Helmet Safety*" — Developed real-time helmet identification using deep learning for player safety in NFL environments. Published on arXiv, 2022.
arxiv.org/abs/2205.09878
- * **Nuthalapati, N.V.K.C.**, et al., "*Matrix Hashing with Two Levels of Collision Resolution*" — Proposed a novel hashing technique improving collision management efficiency. Presented at IEEE Confluence 2018.
ieeexplore.ieee.org/abstract/document/8442607
- * **Nuthalapati, V.K.C.**, et al., "*An Approach for Misleading Image Outlier Elimination for Smart City Data*" — Improved data integrity and noise filtering for smart-city intelligence systems. Published by Springer LNCS, 2018.
link.springer.com/.../978-981-10-7868-2_24